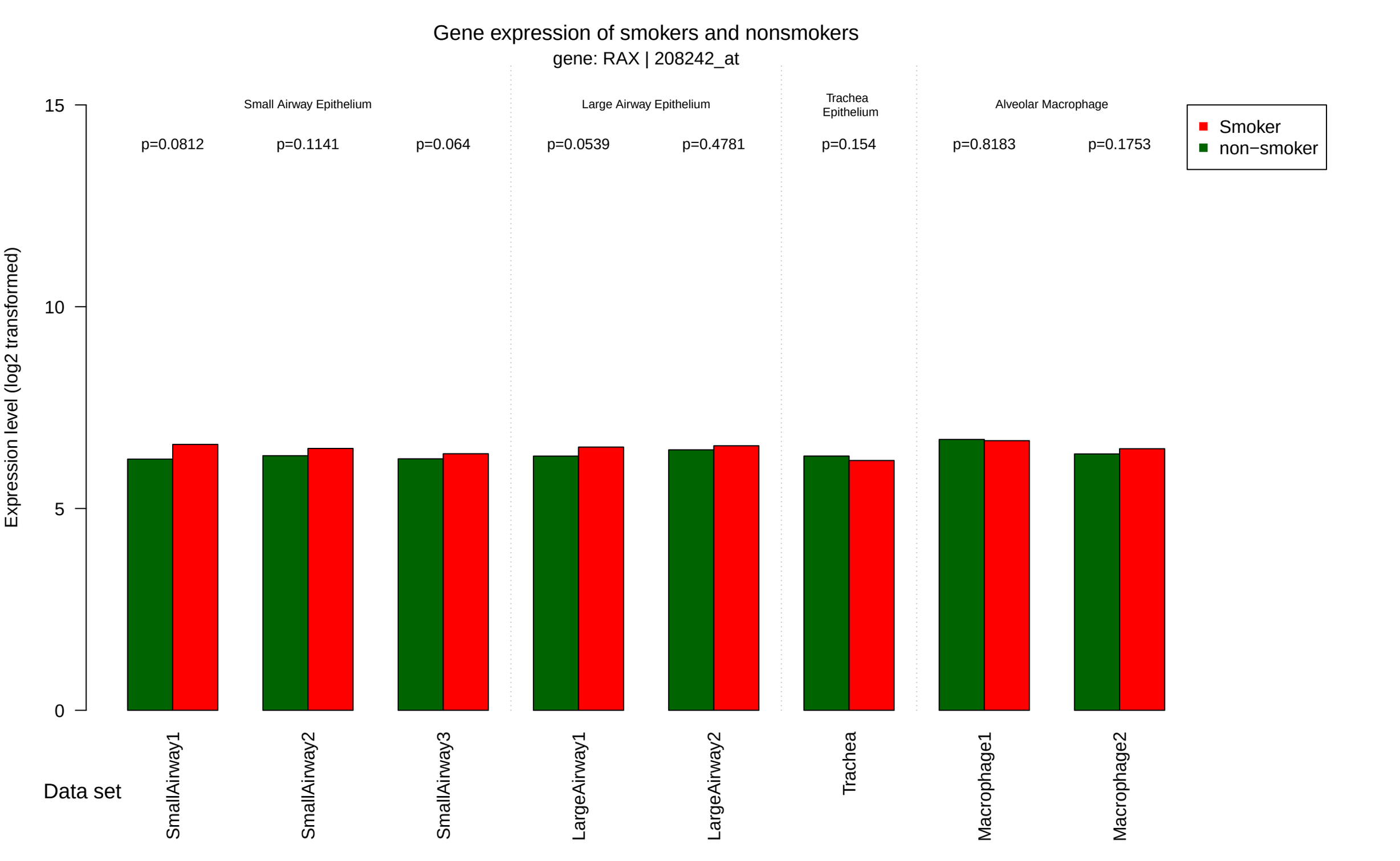
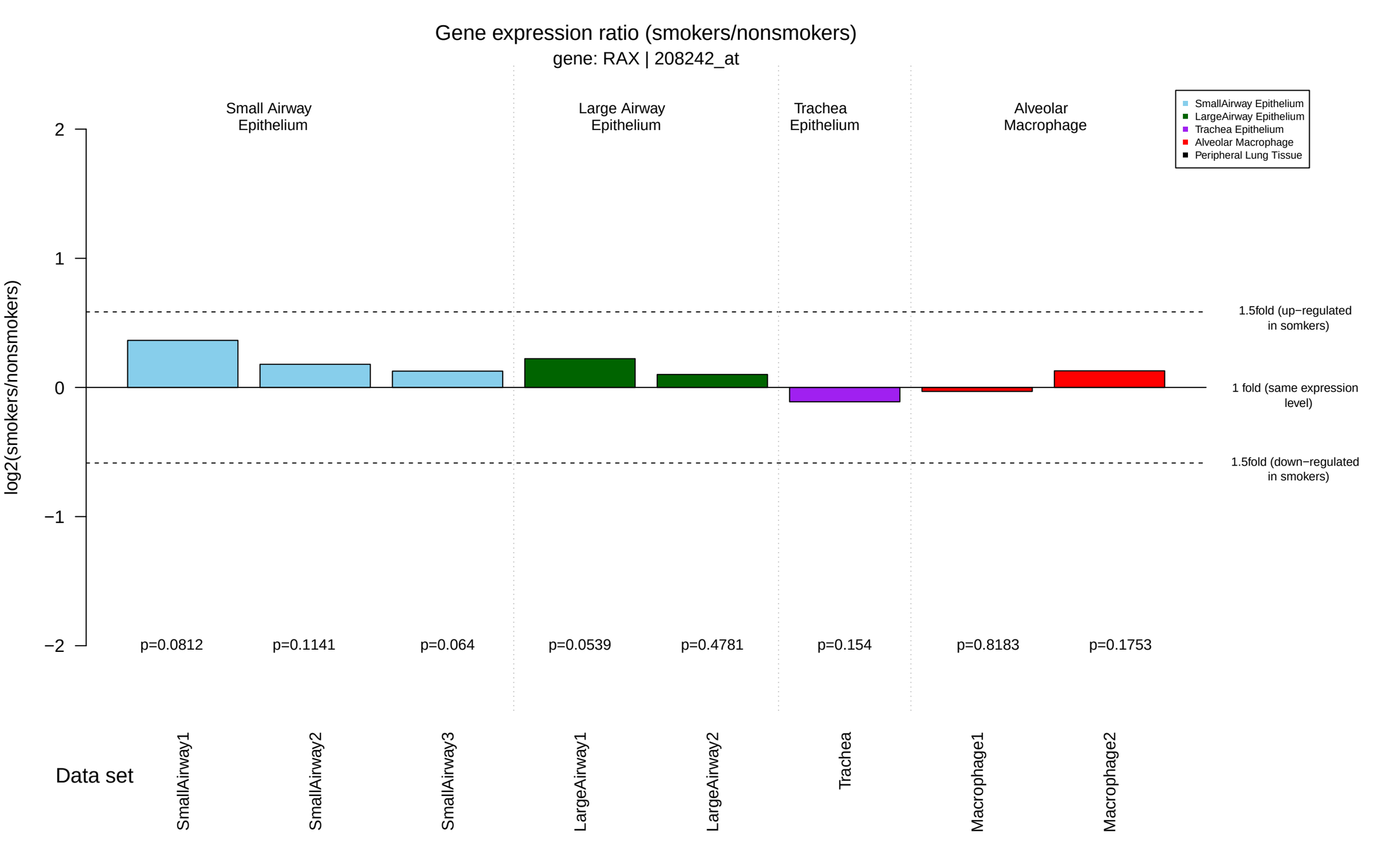
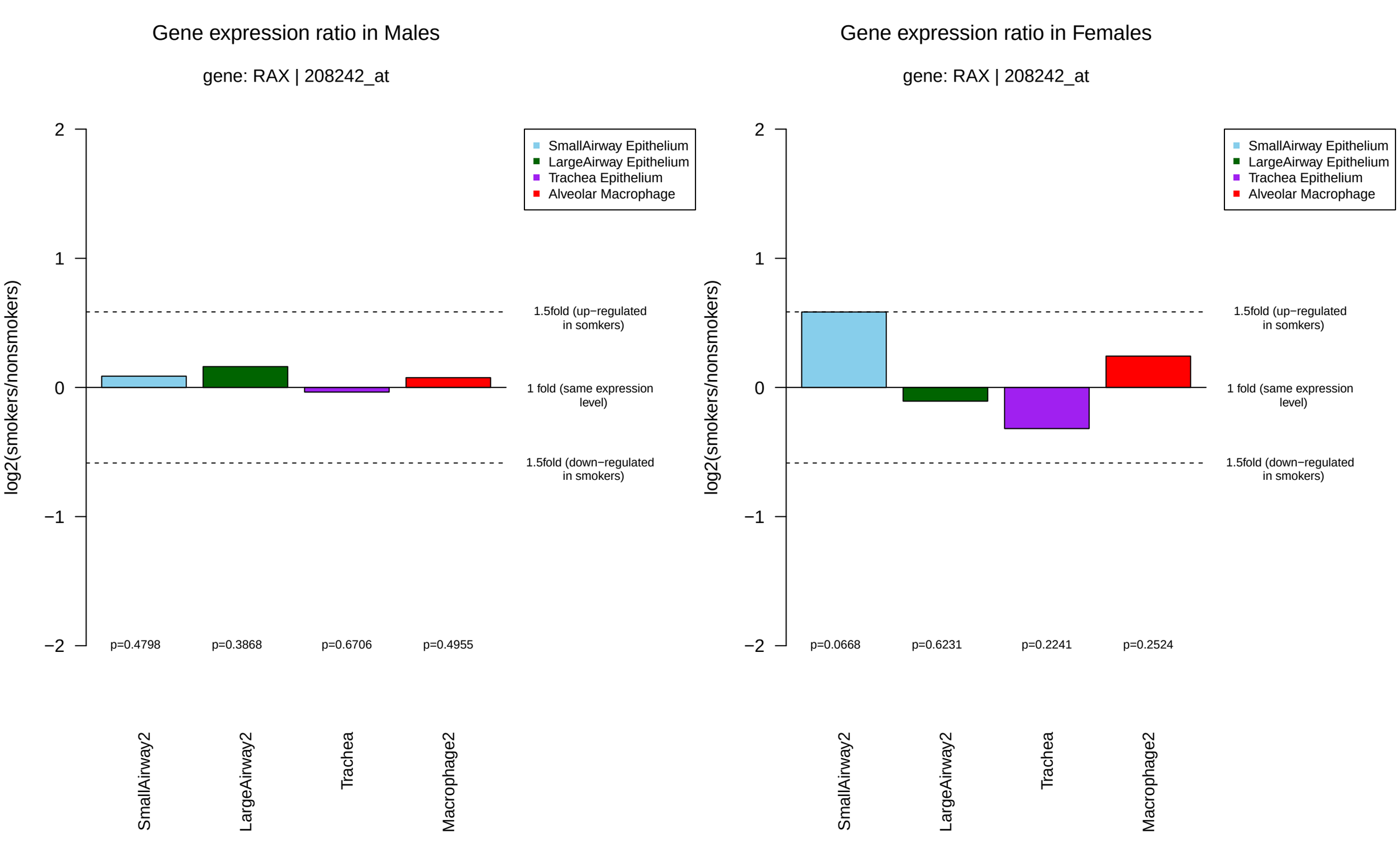
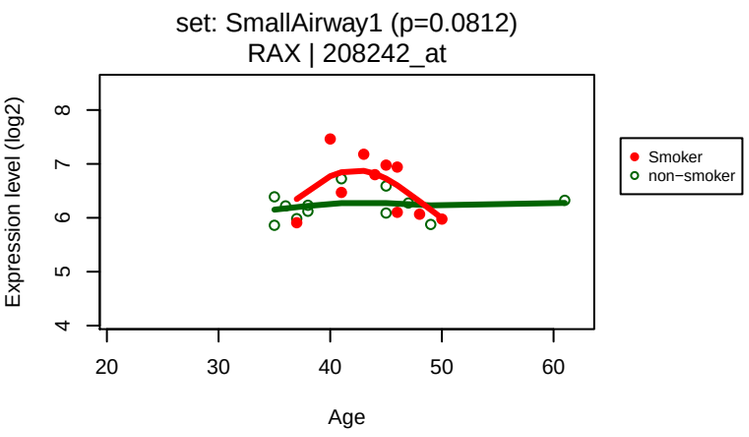
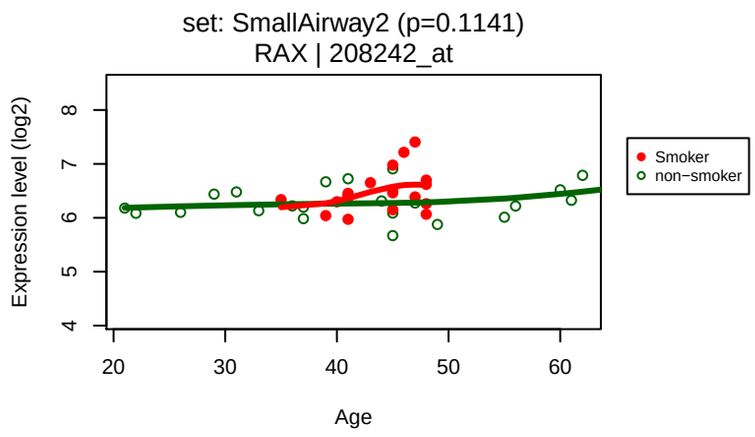
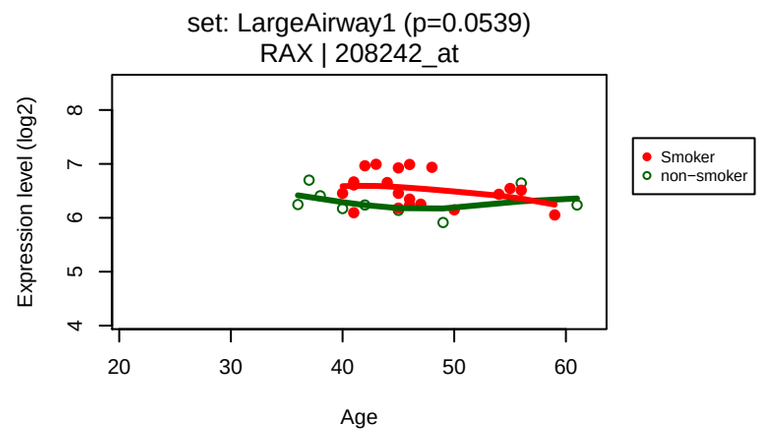

Gene expression of smokers and nonsmokers
gene: RAX | 208242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0812
p=0.1141
p=0.064
p=0.0539
p=0.4781
p=0.154
p=0.8183
p=0.1753
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

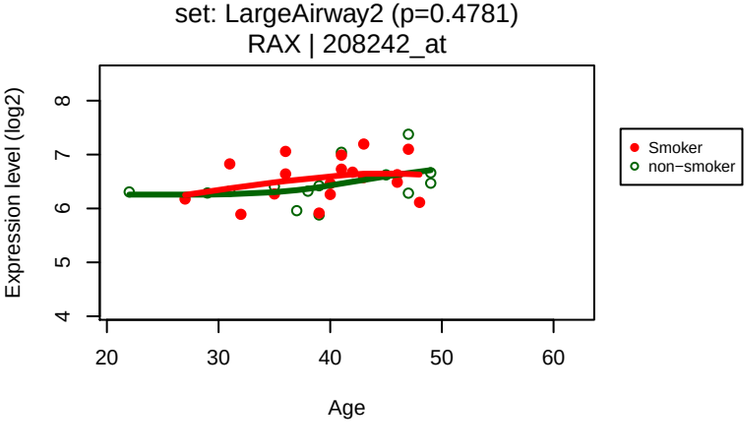

Gene expression ratio (smokers/nonsmokers)
gene: RAX | 208242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0812
p=0.1141
p=0.064
p=0.0539
p=0.4781
p=0.154
p=0.8183
p=0.1753
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

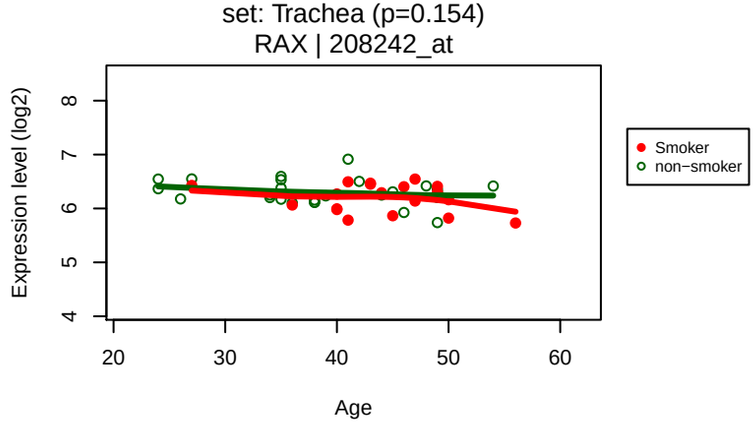

Gene expression ratio in Males
gene: RAX | 208242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4798
p=0.3868
p=0.6706
p=0.4955
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAX | 208242_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0668
p=0.6231
p=0.2241
p=0.2524
SmallAirway2
LargeAirway2
Trachea
Macrophage2

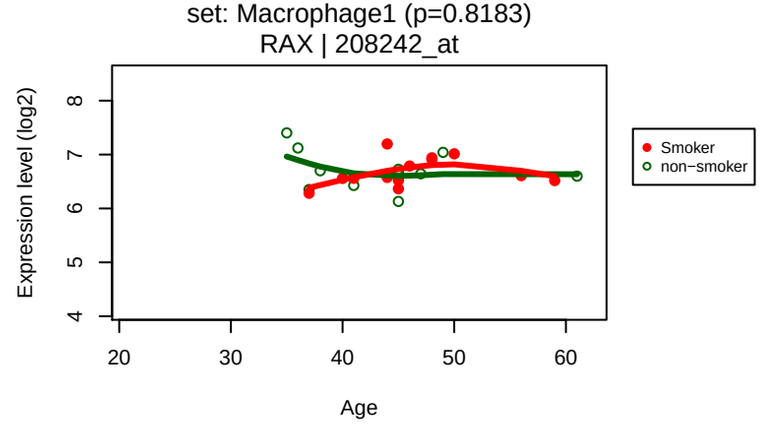

set: SmallAirway1 (p=0.0812)
RAX | 208242_at
Expression level (log2)
Age
Smoker
non−smoker

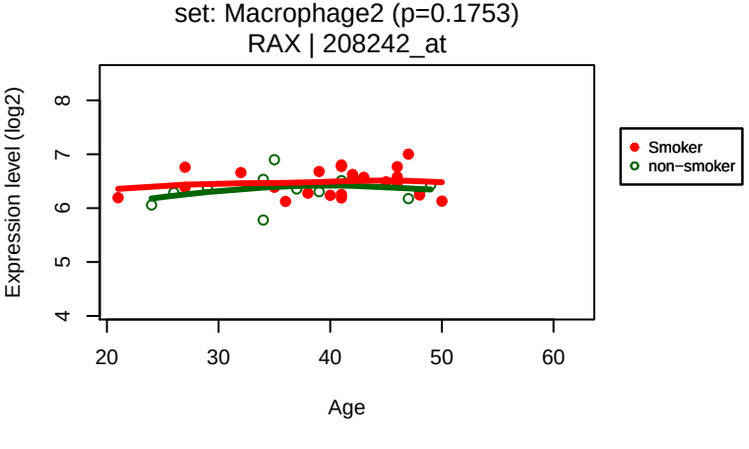

set: SmallAirway2 (p=0.1141)
RAX | 208242_at
Expression level (log2)
Age
Smoker
non−smoker

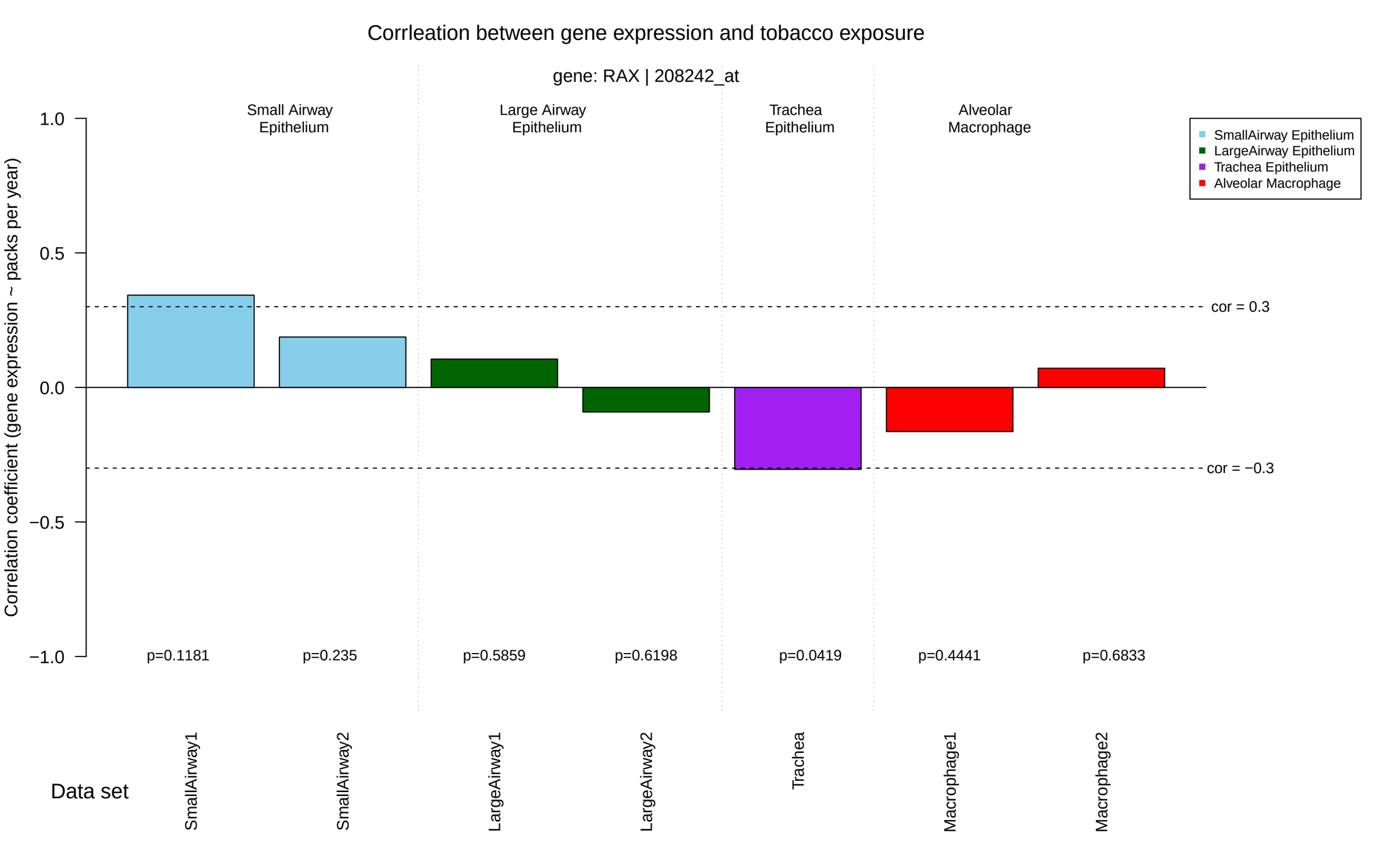

set: LargeAirway1 (p=0.0539)
RAX | 208242_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4781)
RAX | 208242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.154)
RAX | 208242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8183)
RAX | 208242_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1753)
RAX | 208242_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAX | 208242_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1181
p=0.235
p=0.5859
p=0.6198
p=0.0419
p=0.4441
p=0.6833
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2